

BANKING RISK ANALYTICS

Project Report

Lending Risk Overview & Customer Exposure Analysis

Prepared by

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Dataset	Banking.csv - 3,000 customer records
Technology Stack	Python, Pandas, NumPy, Matplotlib, MySQL, Power BI, DAX, Power Query
Project Focus	Risk segmentation, lending exposure, financial strength, branch analysis, and Customer 360

Executive Summary

This project develops an end-to-end banking risk analytics solution that transforms customer banking data into actionable lending and portfolio insights. Python is used for data inspection, preprocessing, feature engineering, and exploratory data analysis; MySQL supports structured querying and aggregation; and Power BI provides a six-page interactive dashboard for risk overview, customer segmentation, lending exposure, financial strength, branch comparison, and Customer 360 decision support.

Portfolio Snapshot

The dataset contains 3,000 customer records and 25 source columns. It has no missing values and no exact duplicate rows. There are 3,000 unique Client IDs, so every row represents a unique customer. Total bank loans are ₹1.77 B, total bank deposits are ₹2.01 B, business lending is ₹2.60 B, and the portfolio loan-to-deposit ratio is 88.06%.

Key KPI	Value
Total Customers	3,000
Unique Client IDs	3,000
Total Bank Loans	₹1.77 B
Total Bank Deposits	₹2.01 B
Total Business Lending	₹2.60 B
Average Risk Weighting	2.25
Risk Weighting 4-5	482 (16.07%)
Loan-to-Deposit Ratio	88.06%
Branches	4

Report Structure

1. Project overview, problem statement, objectives, and workflow
2. Dataset profile, data quality checks, and preprocessing
3. Feature engineering and exploratory data analysis
4. Risk, lending, financial strength, and branch analysis
5. MySQL integration and SQL operations
6. Power BI dashboard architecture and DAX measures
7. Key insights, recommendations, limitations, and conclusion

1. Project Overview

Banks and financial institutions need to understand customer financial capacity, existing exposure, deposit strength, and portfolio concentration before making lending decisions. This project organizes those variables into an analytical workflow that supports monitoring and comparison rather than automated credit approval.

1.1 Problem Statement

Develop a practical understanding of risk analytics in banking and financial services, and demonstrate how customer banking data can be used to reduce the risk of financial loss while lending by identifying exposure patterns, higher-risk segments, and customers requiring closer review.

1.2 Objectives

- Clean and validate the banking dataset before analysis.
- Segment customers by income, age, loyalty, fee structure, branch, and risk weighting.
- Compare bank loans, bank deposits, business lending, and credit-card exposure.
- Measure portfolio-level Loan-to-Deposit Ratio and related financial-strength indicators.
- Compare branches using deposits, loans, customer counts, and risk mix.
- Build a Customer 360 view for single-customer financial and lending review.
- Create an interactive Power BI dashboard and reusable SQL analytical layer.

1.3 End-to-End Workflow



Figure 1. End-to-end analytics workflow used in the project.

2. Dataset Description and Initial Inspection

The source file contains 3,000 rows and 25 columns. Each row represents one unique customer record with demographic, relationship, account, lending, deposit, and risk-related attributes.

2.1 Main Data Fields

Category	Fields
Client / Profile	Client ID, Name, Age, Nationality, Occupation, GenderId
Relationship	Joined Bank, Banking Contact, Loyalty Classification, Fee Structure
Income / Wealth	Estimated Income, Superannuation Savings, Properties Owned
Credit / Lending	Bank Loans, Business Lending, Credit Card Balance, Amount of Credit Cards
Deposits / Accounts	Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account
Risk / Organization	Risk Weighting, BRId, Location ID, IAId

2.2 Initial Inspection Results

- Dataset shape: 3,000 rows x 25 columns.
- Missing values: 0.
- Exact duplicate rows: 0.
- Unique Client IDs: 3,000; every customer record has a unique Client ID.
- Age range: 17 to 85, with an average age of 51.0.
- Average Estimated Income: ₹171.31 K; median: ₹142.31 K.

Primary-Key Design Decision

The corrected dataset contains 3,000 unique Client IDs. Client ID can therefore be used as the customer-level business key for Power BI drill-through and SQL lookups. A surrogate row_id may still be retained in the database implementation if desired.

3. Data Quality Checks and Preprocessing

3.1 Data Quality Validation

- Checked missing values across all columns using pandas null-value summaries.
- Checked exact duplicate rows and found none.
- Reviewed data types and numerical ranges for major financial variables.
- Verified category values for Risk Weighting, BRId, Loyalty Classification, Fee Structure, Nationality, and GenderId.
- Verified Client ID uniqueness: all 3,000 customer records have distinct Client IDs.

3.2 Data Transformation

- Converted the Joined Bank field to a date-compatible format for database and reporting use.
- Standardized analytical labels for income bands, age groups, risk categories, and branch names.
- Created SQL-friendly snake_case field names in the MySQL analytical layer.
- Used display measures in Power BI to present monetary values in rupees, millions, and billions without changing underlying values.

3.3 Data-Modeling Cautions

- Saving Accounts and Checking Accounts appear to behave as monetary variables in the dataset; they should not be treated as account counts unless source documentation confirms that meaning.
- Bank Deposits may overlap conceptually with Saving Accounts and Checking Accounts. Therefore, any combined “Total Account Balance” is treated as a project-defined analytical metric, not an audited balance-sheet measure.
- Higher Risk Weighting values are used for segmentation in this project, but the dataset does not provide a formal regulatory definition of the scale.

3.4 Python Libraries

The EDA notebook uses Python with Pandas and NumPy for data handling and Matplotlib/Seaborn for visualization. MySQL connectivity is performed with mysql.connector / SQL interfaces in the notebook workflow.

4. Feature Engineering

4.1 Income Band

Estimated Income is segmented into three project bands to support comparison across customer financial capacity levels:

Income Band	Rule	Customers	Share
Low	Estimated Income < ₹100,000	1,027	34.2%
Medium	₹100,000 <= Income < ₹300,000	1,517	50.6%
High	Estimated Income >= ₹300,000	456	15.2%

4.2 Age Group

Age Group	Customers
Under 30	564
30-39	431
40-49	451
50-59	427
60+	1,127

4.3 Derived Analytical Measures

- Loan-to-Deposit Ratio = Total Bank Loans / Total Bank Deposits.
- Risk Weighting 4-5 Customers = count of customers with Risk Weighting >= 4.
- Total Credit Exposure = Bank Loans + Business Lending + Credit Card Balance.
- Branch Name = “Branch ” + BRId.
- Risk Category = “Risk ” + Risk Weighting.

5. Exploratory Data Analysis (EDA)

EDA was performed to understand category distributions, portfolio concentration, relationships among financial variables, and patterns that can later be used in SQL and Power BI.

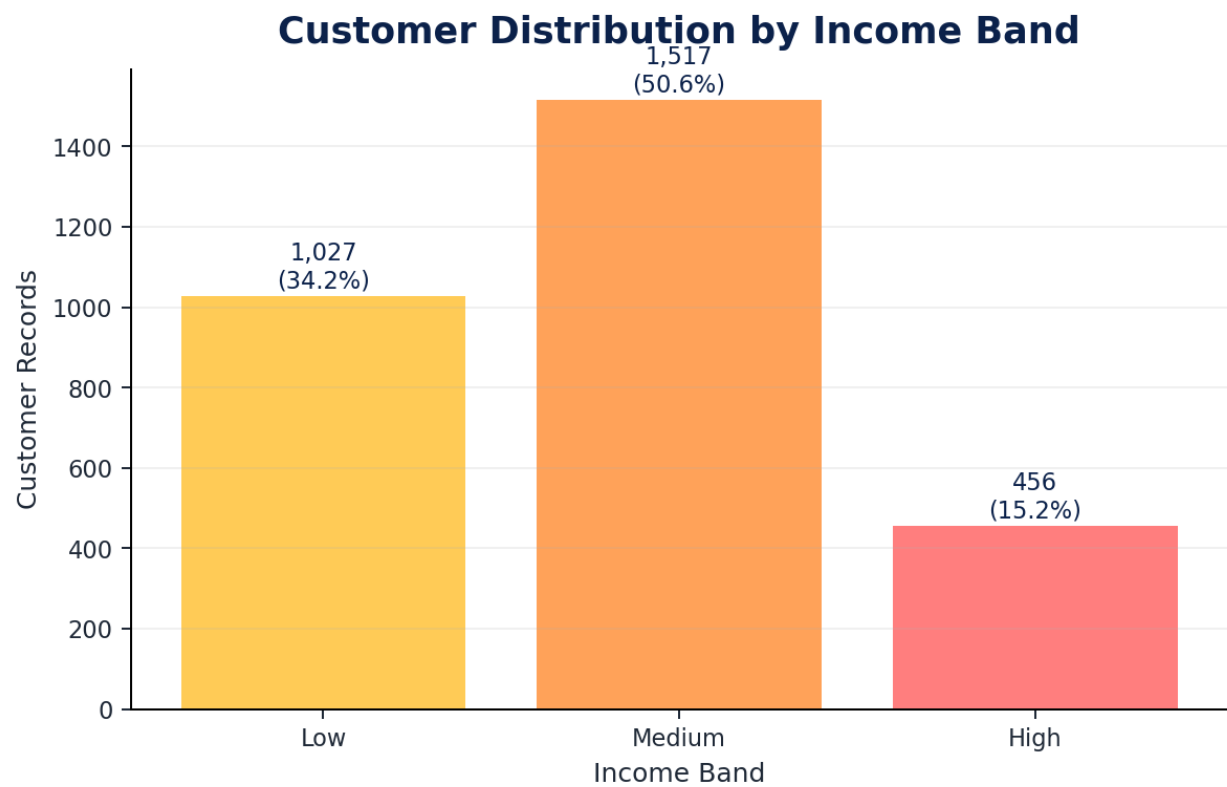


Figure 2. Customers by Income Band.

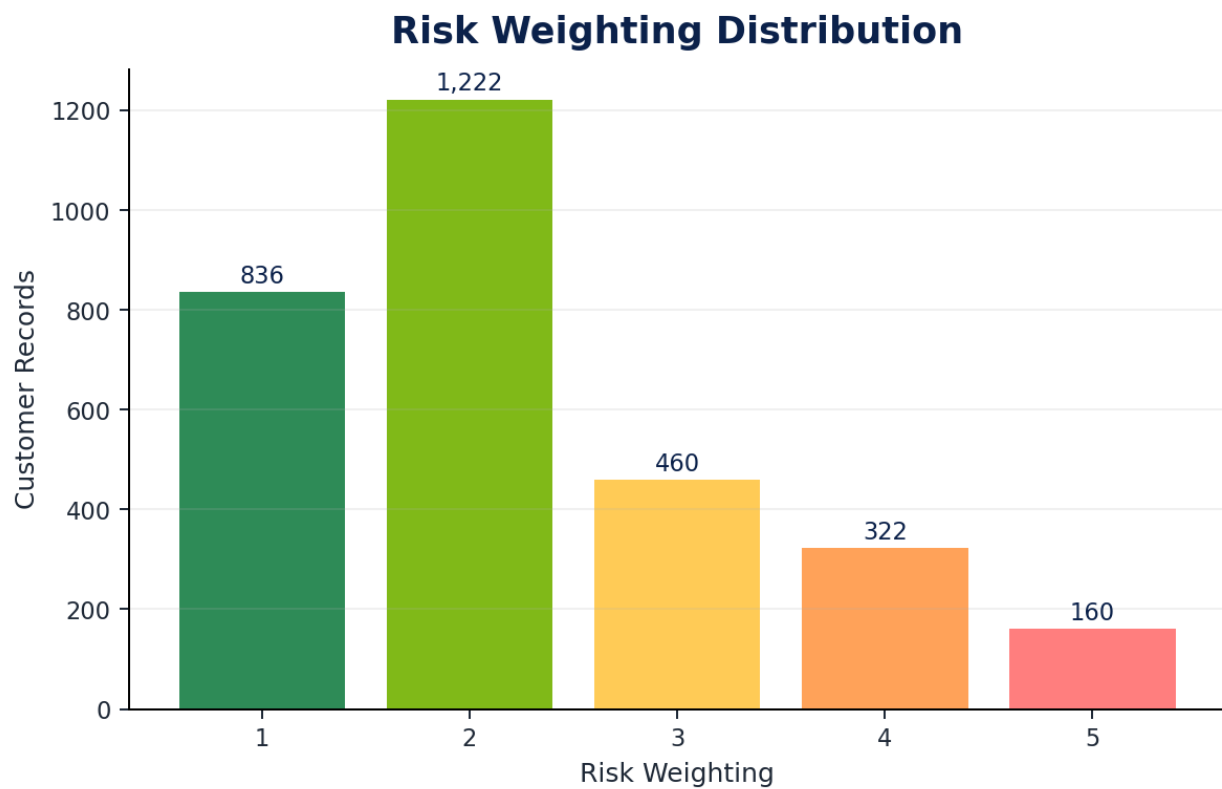


Figure 3. Risk Weighting distribution.

5.1 EDA Observations

- Medium-income customers are the largest segment with 1,517 customers (50.6%).
- Risk Weighting 2 is the most common category with 1,222 customers (40.7%).
- Risk Weighting 4-5 together account for 482 customers (16.07%).
- Branch 3 contains 1,352 customers and is the largest branch segment in the dataset.

6. Numerical Relationship and Correlation Analysis

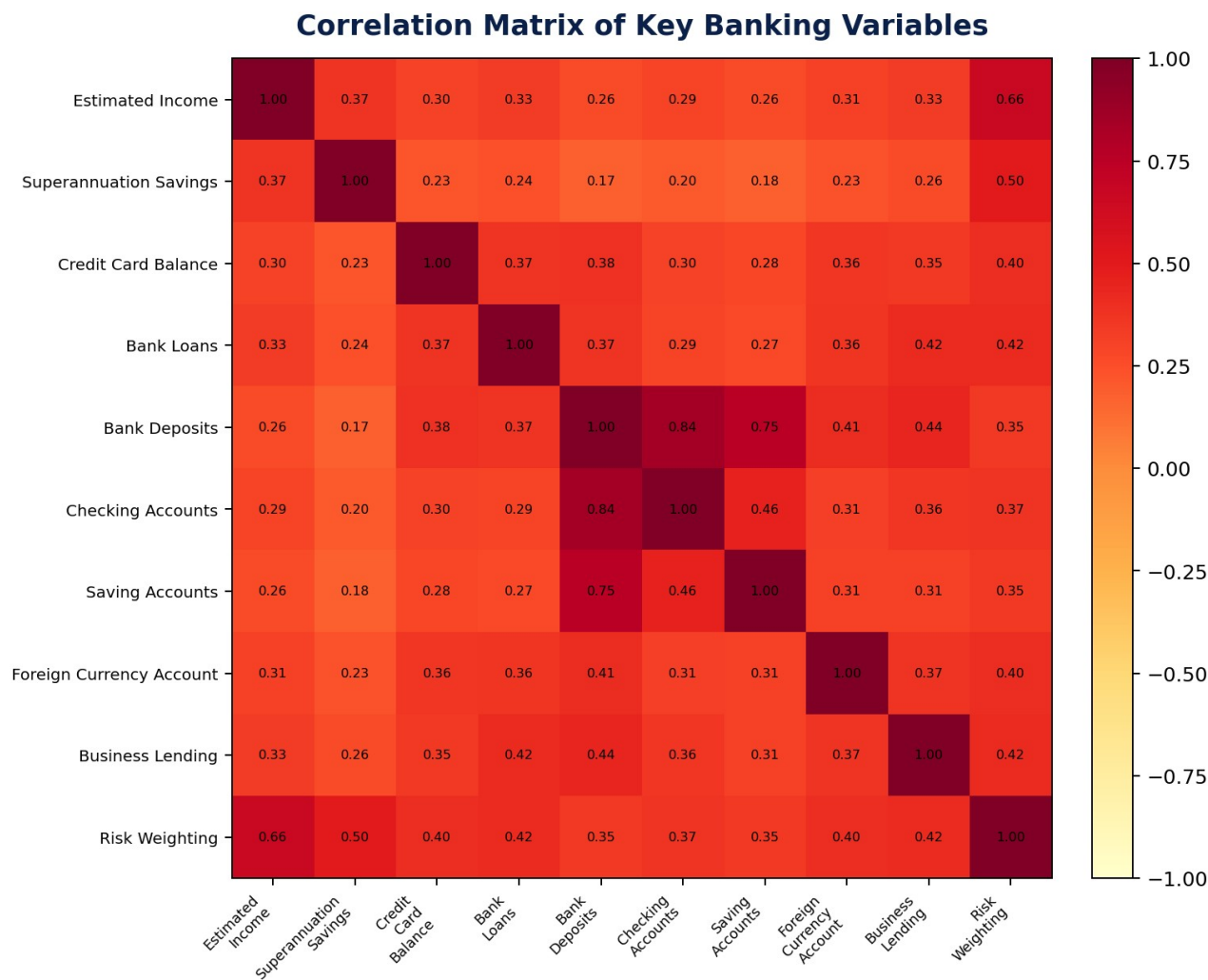


Figure 4. Correlation matrix of selected numerical variables.

6.1 Key Relationship Findings

- Bank Deposits has a strong positive correlation with Checking Accounts (approximately 0.84) and Saving Accounts (approximately 0.75), showing that deposit strength moves closely with account balances in this dataset.
- Estimated Income has a moderate positive correlation with Risk Weighting (approximately 0.66); this is a data relationship, not a causal or policy conclusion.
- Bank Loans and Business Lending show a moderate positive relationship (approximately 0.42), indicating that some customers carry both personal and business borrowing exposure.
- Properties Owned has comparatively weak relationships with several banking variables, suggesting property ownership may depend on factors outside the available dataset.

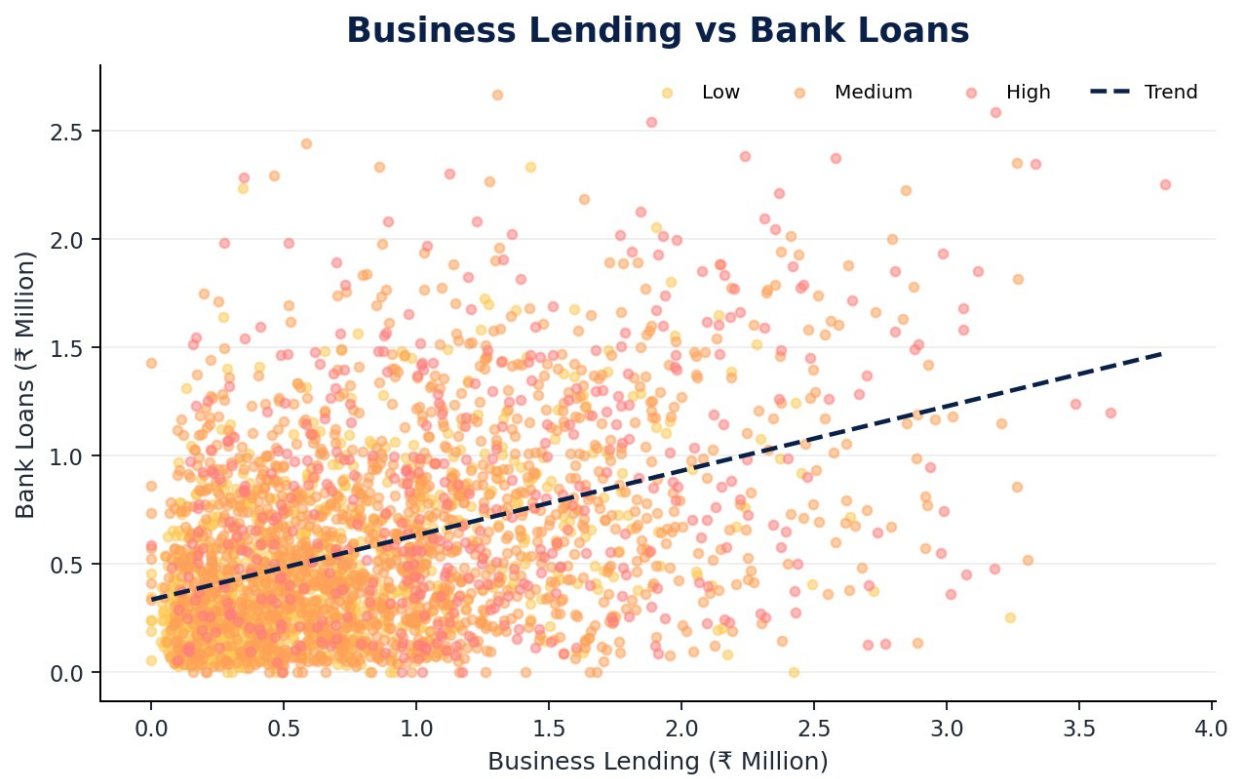


Figure 5. Business Lending versus Bank Loans, segmented by Income Band.

7. Customer Risk Analysis

Risk segmentation is used to compare the distribution of customers across income, loyalty, fee structure, age, nationality, gender, and branch dimensions. The dashboard keeps the five Risk Weighting categories visible so that portfolio composition can be examined without collapsing all observations into a single label.

Risk Weighting	Customers	Share
Risk 1	836	27.9%
Risk 2	1,222	40.7%
Risk 3	460	15.3%
Risk 4	322	10.7%
Risk 5	160	5.3%

Risk 4-5 Monitoring Segment

482 customers fall in Risk Weighting 4 or 5, representing 16.07% of the dataset. The project uses this as a monitoring segment for comparative analytics, not as a regulatory default classification.

7.1 Recommended Segmentation Views

- Risk distribution by Income Band.
- Risk distribution by Loyalty Classification.
- Risk distribution by Fee Structure.
- Risk distribution by Age Group and Nationality.
- Risk mix by Branch.

8. Lending Exposure Analysis

Lending KPI	Value
Total Bank Loans	₹1.77 B
Total Business Lending	₹2.60 B
Average Bank Loan	₹591.39 K
Loan-to-Deposit Ratio	88.06%
Risk 4-5 Loan Exposure	₹429.31 M

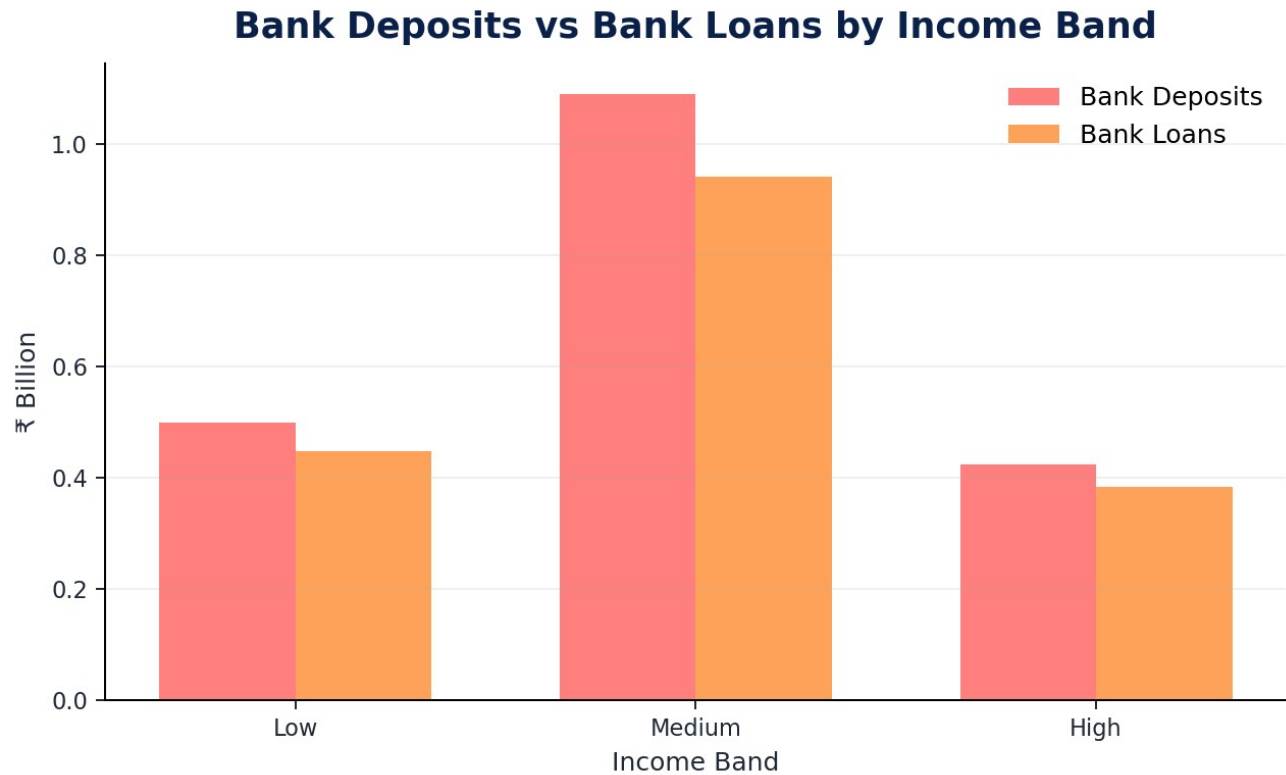


Figure 6. Bank Deposits versus Bank Loans by Income Band.

8.1 Lending Interpretation

- At the portfolio level, loans equal 88.06% of deposits, so total deposits exceed total bank loans.
- Business Lending is larger than Bank Loans at the aggregate level, showing that business exposure is a major part of this dataset.
- The income-band view helps compare whether borrowing and deposit strength scale together across Low, Medium, and High groups.
- Top-customer and Risk Weighting views help identify concentrated exposure for closer review.

9. Financial Strength and Stability Analysis

The financial-strength page compares deposits, loans, estimated income, account balances, and coverage. Coverage Ratio in this project is defined as Total Bank Deposits divided by Total Bank Loans; it is an analytical ratio for this dashboard, not a regulatory capital or liquidity ratio.

Financial KPI	Value
Total Bank Deposits	₹2.01 B
Total Bank Loans	₹1.77 B
Deposit-to-Loan Coverage	113.56%
Average Estimated Income	₹171.31 K
Total Credit Card Balance	₹9.53 M

9.1 Project Financial-Strength Status

For dashboard storytelling, an optional project-defined status can classify coverage ≥ 1.00 as Strong, 0.85-0.99 as Moderate, and below 0.85 as Review. These thresholds are internal analytical labels only and should not be presented as banking regulation.

9.2 Customer-Level Measures

- Selected Estimated Income
- Selected Bank Deposits
- Selected Bank Loans
- Selected Business Lending
- Selected Credit Card Balance
- Customer Loan-to-Deposit Ratio
- Customer Credit Exposure

10. Branch Performance and Comparative Analysis

Branch	Customers	Deposits	Loans	L/D Ratio	Avg Risk	Risk 4-5 %
Branch 1	660	₹425.49 M	₹380.88 M	89.52%	2.22	16.1%
Branch 2	495	₹328.97 M	₹278.60 M	84.69%	2.31	18.0%
Branch 3	1,352	₹925.40 M	₹814.22 M	87.99%	2.23	15.1%
Branch 4	493	₹334.82 M	₹300.47 M	89.74%	2.29	16.8%

Branch Portfolio Comparison

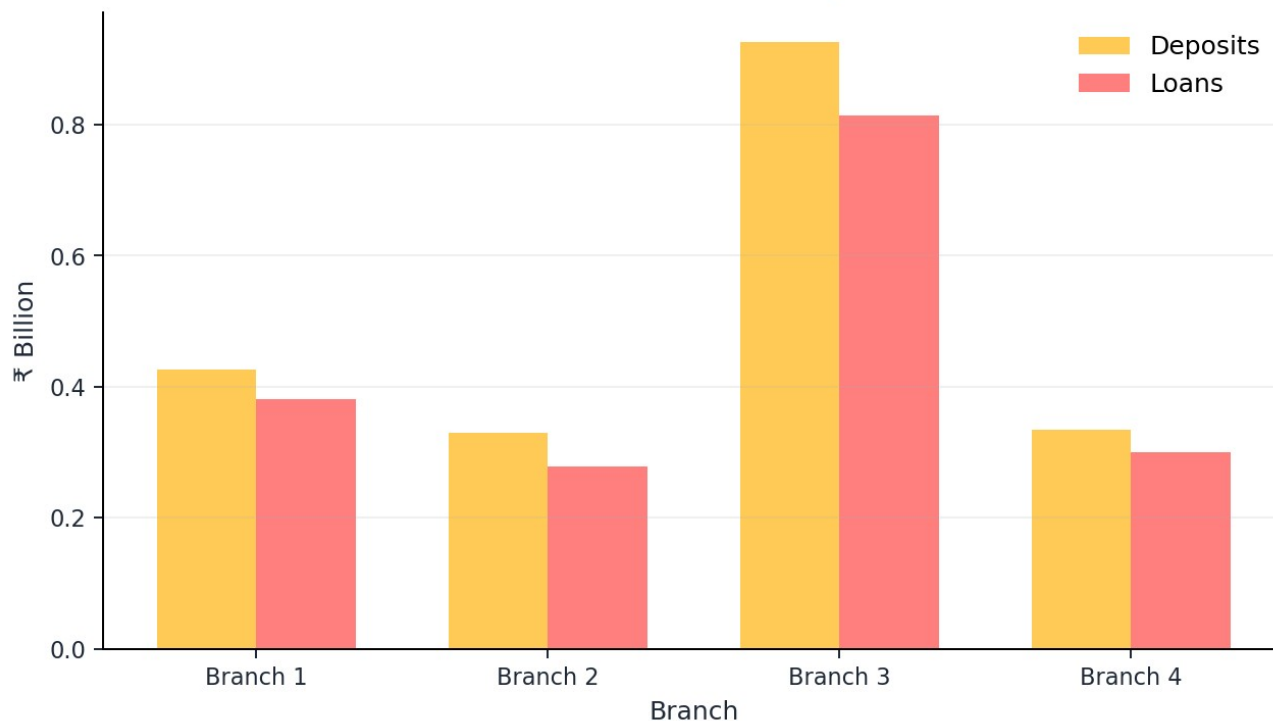


Figure 7. Branch-level deposits and loans.

10.1 Branch Findings

- Branch 3 is the largest by customer count (1,352).
- Branch 4 has the highest loan-to-deposit ratio at 89.74%, closely followed by Branch 1.
- Branch 2 has the highest Risk Weighting 4-5 share at 18.0%.
- Branch comparison is useful for prioritizing portfolio review, but differences should be interpreted with branch size and customer mix in mind.

11. MySQL Database Integration and SQL Analysis

The project includes a MySQL analytical layer for repeatable queries, validation, aggregation, and integration with reporting tools. A self-contained SQL operations file was created for the dataset with table creation, row-level inserts, indexes, a derived analytical view, data-quality checks, dashboard queries, and CRUD examples.

11.1 Database Design

- Database: banking_risk_analytics
- Primary table: banking_customer
- Unique business key: client_id; an optional row_id may be retained as a surrogate database key
- Analytical view: vw_banking_analytics
- Indexes on Client ID, Branch ID, Risk Weighting, Nationality, Loyalty, Fee Structure, and Joined Bank

11.2 Example SQL - Portfolio KPIs

```
SELECT
  COUNT(*) AS total_customers,
  SUM(bank_loans) AS total_bank_loans,
  SUM(bank_deposits) AS total_bank_deposits,
  AVG(risk_weighting) AS avg_risk_weighting,
  100.0 * SUM(bank_loans) / NULLIF(SUM(bank_deposits),0) AS loan_to_deposit_ratio_pct
FROM banking_customer;
```

11.3 Example SQL - Risk Distribution

```
SELECT risk_weighting, COUNT(*) AS customer_count
FROM banking_customer
GROUP BY risk_weighting
ORDER BY risk_weighting;
```

11.4 SQL Analysis Areas

- Portfolio KPIs and risk distribution
- Income-band and lending segmentation
- Top customers by loan amount and total exposure
- Branch performance summary and risk mix
- Customer 360 lookup by Client ID
- Project-defined lending status and recommended-action logic

12. Power BI Dashboard

The Power BI report is organized as a six-page navigation experience using a consistent cream, yellow, orange, coral, and navy theme. Slicers, page navigation, reset-filter bookmarks, and drill-through are used to support interactive analysis.

Page	Purpose	Main Content
1. Risk Overview	Portfolio-level lending and risk snapshot	Customers, loans, deposits, average risk, Risk 4-5, L/D ratio, risk distribution, income-band analysis
2. Customer Risk	Customer segmentation and profile risk	Risk by income, loyalty, fee structure, age, nationality, and gender
3. Lending Exposure	Borrowing and portfolio exposure	Loans, business lending, L/D ratio, Risk 4-5 exposure, top borrowers, portfolio summary
4. Financial Strength	Deposit coverage and customer financial capacity	Deposits, loans, coverage, income, account balance, strength summary
5. Branch Analysis	Branch comparison	Customers, deposits, loans, L/D ratio, risk mix, loyalty mix, branch summary
6. Customer 360	Single-customer lending decision support	Profile, selected financial metrics, exposure, L/D gauge, product holdings, insight, recommended action

12.1 Interactive Features

- Page Navigator for movement across all six report pages.
- Slicers for Income Band, Risk Weighting, BRId, Loyalty Classification, Fee Structure, Age Group, Nationality, and GenderId where appropriate.
- Reset Filters button implemented with a bookmark whose Data setting is enabled.
- Customer 360 can use the unique Client ID for reliable single-customer drill-through.
- Consistent Risk 1-5 legend order and colors across risk visuals.

12.2 Theme Palette

Use	Color
Page background	#FFF9EF
Header cream	#FFEDB9
Golden yellow	#FFCB56
Orange	#FFA259
Coral	#FF7E7E
Dark navy text	#0B214A
White cards	#FFFFFF

13. Core DAX Measures

Total Customers

Total Customers = COUNTROWS('Banking')

Total Bank Loans

Total Bank Loans = SUM('Banking'[Bank Loans])

Total Bank Deposits

Total Bank Deposits = SUM('Banking'[Bank Deposits])

Loan to Deposit Ratio

Loan to Deposit Ratio =

```
DIVIDE(  
    [Total Bank Loans],  
    [Total Bank Deposits],  
    0  
)
```

Risk 4-5 Customers

Risk 4-5 Customers =

```
CALCULATE(  
    COUNTROWS('Banking'),  
    'Banking'[Risk Weighting] >= 4  
)
```

Risk 4-5 %

Risk 4-5 % =

```
DIVIDE(  
    [Risk 4-5 Customers],  
    [Total Customers],  
    0  
)
```

Coverage Ratio

Coverage Ratio =

```
DIVIDE(  
    [Total Bank Deposits],  
    [Total Bank Loans],  
    0  
)
```

14. Customer 360 Decision-Support Logic

The Customer 360 page combines a selected customer profile with financial measures, risk weighting, product holdings, and a project-defined recommended action. It is designed to support analyst review rather than to make an automated approval decision.

14.1 Customer Risk Insight Logic

```
Customer Risk Insight =  
VAR RiskScore = MAX('Banking'[Risk Weighting])  
VAR Ratio = [Customer Loan to Deposit Ratio]  
RETURN  
SWITCH(  
    TRUE(),  
    RiskScore >= 4 && Ratio > 1, "Higher risk weighting and loans above deposits indicate a customer requiring enhanced lending review.",  
    RiskScore >= 4, "The customer is in the upper risk-weighting range and should be monitored closely before additional lending.",  
    RiskScore <= 2 && Ratio < 0.75, "The customer has lower risk weighting with relatively strong deposit coverage.",  
    Ratio > 1, "Loan exposure exceeds deposits, indicating comparatively higher leverage.",  
    "The customer shows a moderate financial and risk profile."  
)
```

14.2 Recommended Lending Action

- **PRIORITY REVIEW:** higher Risk Weighting combined with loans above deposits.
- **MONITOR:** higher Risk Weighting or elevated loan-to-deposit ratio.
- **OPPORTUNITY:** lower Risk Weighting with strong deposit coverage.
- **STANDARD MONITORING:** other moderate profiles.

Important

The above thresholds are project-defined business rules for dashboard demonstration. They are not credit policy, underwriting guidance, or regulatory rules.

15. Key Insights

- The corrected portfolio contains 3,000 records and 3,000 unique Client IDs, so each row represents one unique customer.
- Total bank deposits (₹2.01 B) exceed total bank loans (₹1.77 B), producing an overall L/D ratio of 88.06%.
- Medium-income customers represent the largest segment at 50.6% of customers.
- Risk Weighting 2 is the largest category at 40.7%; Risk Weighting 4-5 together represent 16.07%.
- Branch 3 is the largest branch segment with 1,352 customers.
- Bank Deposits is strongly associated with Checking Accounts and Saving Accounts in the numerical correlation analysis.
- Bank Loans and Business Lending move together moderately, supporting the use of combined exposure views at customer level.

16. Recommendations

- Use Client ID throughout the data model where one-customer identification is required.
- Monitor Risk Weighting 4-5 segments alongside actual exposure amounts instead of relying on count alone.
- Review customers with high total credit exposure and elevated loan-to-deposit ratios before additional lending.
- Compare branch L/D ratio and Risk 4-5 share together so branch size and portfolio mix are both considered.
- Keep risk, financial-strength, and lending-status thresholds documented as project-specific unless formal business policy is available.
- Validate the semantic meaning of Bank Deposits, Saving Accounts, and Checking Accounts before using additive “total balance” metrics in production.
- Automate data refresh and SQL-to-Power BI connectivity if the project is extended to live or periodic data.

17. Limitations and Future Enhancements

17.1 Current Limitations

- The dataset does not provide an explicit target variable such as loan default, delinquency, or loss outcome, so the project performs descriptive risk analytics rather than predictive credit-risk modeling.
- Risk Weighting definitions are not documented in the source dataset.
- The corrected dataset contains a unique Client ID for each of the 3,000 customer records.
- Some account/deposit fields may overlap semantically and require a source data dictionary for production interpretation.
- The dashboard thresholds for Review, Monitor, Strong, or Opportunity are analytical rules created for the project.

17.2 Future Enhancements

- Add loan repayment/default history and build supervised credit-risk models.
- Create time-series monitoring for exposure and deposit trends.
- Add explainable risk scoring using SHAP or interpretable models when a target variable is available.
- Implement scheduled ETL, data validation, and refresh pipelines.
- Add row-level security for branch-specific users.
- Create automated PDF/Excel portfolio reports for management review.

18. Conclusion

The Banking Risk Analytics project demonstrates a complete analytics workflow from raw banking records to decision-support dashboards. Python enables data inspection, preprocessing, feature engineering, and EDA; MySQL provides a structured analytical layer; and Power BI converts portfolio measures into interactive views for risk segmentation, lending exposure, financial strength, branch comparison, and Customer 360 review. The project provides a strong foundation for more advanced credit-risk modeling once verified target outcomes and formal policy definitions are available.

Final Outcome

A reusable end-to-end banking analytics solution combining Python, SQL, and Power BI to support data-driven portfolio monitoring and lending review.

Appendix A - Selected Portfolio Values

Metric	Value
Customers	3,000
Unique Client IDs	3,000
Average Age	51.04
Average Estimated Income	₹171.31 K
Total Bank Loans	₹1.77 B
Total Bank Deposits	₹2.01 B
Total Business Lending	₹2.60 B
Total Credit Card Balance	₹9.53 M
Average Risk Weighting	2.25
Risk 4-5 Customers	482 (16.07%)
Loan-to-Deposit Ratio	88.06%
Number of Branches	4

Appendix B - Income Band Ordering

Income Band Ordered =

```
SWITCH(  
  TRUE(),  
  'Banking'[Estimated Income] < 100000, "01 Low",  
  'Banking'[Estimated Income] < 300000, "02 Medium",  
  "03 High"  
)
```